K L Deemed to be University	Department of Artificial Intelligence and Data Science -- KLAZIZ Course Handout 2025-2026, Even Sem

Course Title	**: VISUAL RECOGNITION AND SCENE UNDERSTANDING**
Course Code	**: 23SDCS19A**
L-T-P-S Structure	**: 0-0-6-4**
Pre-requisite	:
Credits	**: 4**
Course Coordinator	**: Sasidhar Kothuru**
Team of Instructors	:
Teaching Associates	:
Syllabus :	Fundamentals of visual recognition, History and development, Importance in various fields, Applications and real-world examples, Basics of image processing techniques, Image enhancement, Image segmentation, Feature extraction. Overview of machine learning and deep learning algorithms for Object detection and recognition methodologies, Feature-based and deep learning-based object recognition, Performance evaluation and metrics. Overview of scene understanding, Semantic segmentation, Scene classification, Contextual reasoning in scene understanding. Introduction to 3D vision and reconstruction, Depth perception and 3D vision, 3D reconstruction techniques, Applications and challenges in 3D scene understanding. Visual perception in robotics, Object manipulation and grasping, Navigation and mapping using visual data. Overview of ethical and social implications, Bias and fairness in visual recognition, Privacy concerns in visual data, Ethical considerations in deploying visual recognition systems.
Text Books :	1 Computer Vision: Algorithms and Applications, Richard Szeliski , 2010, Springer. 2 Deep Learning for Computer Vision, Rajalingappaa Shanmugamani, 2018, Packt Publishing. 3 Robotics, Vision and Control: Fundamental Algorithms in MATLAB, Peter Corke, 2011, Springer.
Reference Books :	Multiple View Geometry in Computer Vision, Richard Hartley and Andrew Zisserman, 2003, Cambridge. Scene Reconstruction, Pose Estimation and Tracking, Reinhard Koch and Luc Van Gool, 2016, Springer.
Web Links :	1.Deep Learning Specialization– Coursera https://www.coursera.org/specializations/deep-learning 2.Computer Vision (University at Buffalo, SUNY) – Coursera https://www.coursera.org/specializations/computer-vision-cu? 3.Visual Recognition-IIT-Kanpur https://www.cse.iitk.ac.in/users/vinaypn/teaching/visrec/ 4.Deep Learning for Computer Vision Specialization (University of Michigan) – Coursera https://www.coursera.org/specializations/deep-learning-computer-vision? 5.CS231n – Convolutional Neural Networks for Visual Recognition Free slides, videos, assignments. http://cs231n.stanford.edu/ 6.Deep Learning, New York University https://www.youtube.com/user/GoogleTechTalks/deep_learning_nyu_spring_2014 7.Deep Learning, CMU https://deeplearning.cs.cmu.edu/F25/index.html 8.Deep Learning for Computer Vision-NPTEL https://onlinecourses.nptel.ac.in/noc21_cs93/preview
MOOCS :	1.Deep Learning Specialization– Coursera https://www.coursera.org/specializations/deep-learning 2.Computer Vision (University at Buffalo, SUNY) – Coursera https://www.coursera.org/specializations/computer-vision-cu? 3.Visual Recognition-IIT-Kanpur https://www.cse.iitk.ac.in/users/vinaypn/teaching/visrec/ 4.Deep Learning for Computer Vision Specialization (University of Michigan) – Coursera https://www.coursera.org/specializations/deep-learning-computer-vision? 5.Deep Learning for Computer Vision-NPTEL https://onlinecourses.nptel.ac.in/noc21_cs93/preview

Course Rationale :	This course introduces III B.Tech students to the essential concepts and practical skills of visual recognition and scene understanding, enabling them to process images, detect objects, understand scenes, and work with 3D vision. By learning both classical image processing and modern deep learning techniques, students gain the ability to build intelligent vision systems used in robotics, autonomous vehicles, healthcare, and smart applications. The course also creates awareness of ethical issues such as bias, fairness, and privacy in visual data, preparing students to apply computer vision responsibly in real-world scenarios.
Course Objectives :	To introduce the fundamental principles of visual recognition and image processing techniques used in modern computer vision systems. To equip students with the knowledge of machine learning and deep learning algorithms for object detection, recognition, and performance evaluation. To develop the ability to understand and apply 3D vision concepts such as depth perception, stereo vision, and 3D reconstruction. To enable students to interpret scene understanding tasks including semantic segmentation, scene classification, and contextual reasoning. To create awareness on ethical, social, and privacy concerns related to the deployment of visual recognition technologies.
Global Certifications :	OpenCV Certified Professional Google Professional Machine Learning Engineer Certification AWS Machine Learning Specialty Certification Microsoft Azure AI Engineer Associate Nvidia Deep Learning Institute (DLI) – Computer Vision Certification Fundamentals of Deep Learning for Computer Vision
Industry Specific Tools :	Image Processing & Computer Vision Tools (Industry Standard) OpenCV Most widely used CV library Image enhancement, segmentation, feature extraction Used by Intel, Google, Huawei, Tesla 2. Scikit-Image Python-based image processing Used in research labs & biomedical imaging Deep Learning & Object Detection Tools 3. PyTorch Most preferred DL framework in research & product development Used by Meta, Tesla, OpenAI, Nvidia 4.TensorFlow / Keras Industry-scale deployment Used by Google, Airbnb, DeepMind 5. YOLO Family (YOLOv5, YOLOv7, YOLOv8, YOLO-NAS) Standard for real-time object detection Used in surveillance, retail analytics, autonomous systems Scene Understanding Tools . UNet / SegFormer / DeepLab Semantic segmentation models used in biomedical, satellite, and automotive CV CLIP (OpenAI) Scene recognition & image–text reasoning Used by multimodal AI systems Annotation & Dataset Preparation Tools LabelImg / LabelMe Industry-standard bounding box annotation tools CVAT (Computer Vision Annotation Tool) Used by companies like Intel Supports masks, polygons, 3D annotations 19. Label Studio Supports images, video, text, audio Used in ML data pipelines Deployment & Cloud-Based Vision Tools AWS Rekognition Real-world CV tasks: face detection, scene detection, OCR Google Cloud Vision AI Industry-grade vision APIs Microsoft Azure Computer Vision Scene understanding, OCR, facial emotions Ethical AI & Bias-Detection Tools (For CO4) 23. IBM AI Fairness 360 (AIF360) Measures fairness, bias in models Industry trusted Google What-If Tool Model interpretability Helps evaluate bias & performance FairLearn (Microsoft) Toolkit for responsible AI development
Sector Skill Council :	IT–ITeS Sector Skill Council (IT-ITeS SSC NASSCOM) Most relevant SSC for this course Specializes in: Artificial Intelligence Machine Learning Data Science Computer Vision Deep Learning Cognitive computing AI Ethics Annotation & data processing

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Understand the fundamentals of visual recognition and apply basic image processing techniques including enhancement, segmentation, and feature extraction.	PO2,PO3,PO4,PSO2	3

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO2	Analyze and compare object detection and recognition methodologies using feature-based and deep learning-based techniques.	PSO2,PO1,PO3,PO5	4
CO3	Identify and apply suitable 3D vision and reconstruction techniques, and illustrate their applications and challenges.	PO1,PO4,PO3,PSO1	3
CO4	Analyze the ethical and social implications of visual recognition systems by examining issues related to bias, fairness, privacy, and responsible deployment.	PSO1,PO2,PO6,PO7,PO8,PO11	4

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4
CO1	3	**Btl-2** Identify appropriate preprocessing and enhancement techniques for a given image.	**Btl-3** Apply image enhancement operations such as filtering, smoothing, and contrast adjustment.		
CO2	4	**Btl-2** Compare feature-based and deep learning-based recognition pipelines.	**Btl-3** Implement ML/DL models for detection and recognition tasks.	**Btl-3** Analyze object detection architectures such as Haar, HOG+SVM, YOLO, RCNN.	**Btl-4** Evaluate models using accuracy, precision, recall, F1-score, IoU, mAP.
CO3	3	**Btl-2** Identify appropriate 3D vision techniques for depth estimation or reconstruction	**Btl-3** Apply stereo vision algorithms to compute disparity and depth maps.	**Btl-3** Implement segmentation methods (thresholding, region-based, morphological).	
CO4	4	**Btl-4** Analyze sources of bias and fairness issues in datasets and visual models.	**Btl-4** Evaluate privacy concerns related to image/video data collection and surveillance.		

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome

Lecture Course DELIVERY Plan:
NO Delivery Plan Exists

Lecture Session wise Teaching – Learning Plan

No Session Plans Exists

Tutorial Course DELIVERY Plan:
NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Image Enhancement Techniques	CO1
2	Image Segmentation – Thresholding	CO1
3	Edge Detection & Morphology	CO1
4	Feature Extraction – ORB, SIFT, SURF	CO1
5	HOG Features & Classical ML	CO1
6	Train-Test Split, Metrics	CO1
7	Introduction to CNNs	CO1
8	CNN Training on CIFAR-10 / MNIST	CO1
9	Transfer Learning for Recognition	CO1
10	Object Detection – Haar Cascades	CO1
11	Object Detection – YOLO Pretrained	CO1
12	Object Detection – Custom Data Demo	CO1
13	Semantic Segmentation – UNet	CO1
14	Instance Segmentation – Mask R-CNN	CO2
15	Scene Classification	CO2
16	Scene Graph Generation	CO2
17	Stereo Vision – Depth Estimation	CO2
18	3D Reconstruction Basics	CO2
19	Robotics Vision – SLAM & Object Grasping	CO2
20	Object Detection System Integration	CO2

Tutorial Session no	Topics	CO-Mapping
21	Scene Classification and Annotation	CO2
22	Combining UNet & Mask R-CNN	CO2
23	Depth and segmentation fusion	CO2
24	Lane detection and obstacle detection	CO2
25	Performance Evaluation & Metrics Analysis	CO2
26	Deployment – Export Vision Model	CO3
27	Basic sequential I/O & frame extraction	CO3
28	Frame sampling & temporal down sampling	CO3
29	Deployment – Export Vision Model	CO2
30	Keyframe extraction by histogram difference	CO3
31	Integration of Computer Vision Models with Web Interfaces using Flask	CO3
32	Multi-Label Image Classification with PyTorch	CO3
33	Transfer Learning for Image Classification using Pretrained CNNs (ResNet, VGG)	CO3
34	Optical Flow Estimation using Lucas-Kanade Method	CO3
35	Reinforcement Learning in Vision using Python & Torch	CO3
36	Model Quantization and Deployment using ONNX and TensorRT	CO3
37	Vision Transformers using PyTorch	CO3
38	Self-Supervised Learning for Representation Learning in Vision	CO3
39	Synthetic Medical Image Generation with Diffusion Models (MONAI & PyTorch)	CO3

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	demonstrate Image Enhancement Techniques	3	LTC	--- NOT APPLICABLE ---
30	implement Image Enhancement Techniques	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain Image Segmentation – Thresholding	2	PPT	--- NOT APPLICABLE ---
30	lab experiment practice by students	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Demonstrate Edge Detection & Morphology	3	LTC	--- NOT APPLICABLE ---
30	Implement Edge Detection & Morphology	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Demonstrate Feature Extraction – ORB, SIFT, SURF	4	LTC	--- NOT APPLICABLE ---
30	Implement Feature Extraction – ORB, SIFT, SURF	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Apply classical machine learning algorithms (e.g., SVM, KNN, Random Forest) using extracted HOG features.	3	LTC	--- NOT APPLICABLE ---
30	lab experiment practice by students	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Analyze model performance and interpret evaluation results to select the best-performing model.	4	LTC	--- NOT APPLICABLE ---
30	Implement Train-Test Split, Metrics	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	implement CCNs for Image Enhancement	4	LTC	--- NOT APPLICABLE ---
30	experiment practice by students	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Build and train a Convolutional Neural Network (CNN) using deep learning frameworks like TensorFlow/Keras or PyTorch.	3	LTC	--- NOT APPLICABLE ---
30	Evaluate CNN model performance using metrics such as accuracy, loss curves, and confusion matrix.	4	Talk	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Implement transfer learning workflows to classify new image categories with limited data.	3	LTC	--- NOT APPLICABLE ---
30	Use pre-trained CNN models (e.g., VGG, ResNet, MobileNet, EfficientNet) for feature extraction and fine-tuning.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Apply Haar Cascades for detecting faces and other objects using OpenCV.	3	LTC	--- NOT APPLICABLE ---
30	Impement Haar Cascade–based object detection techniques using OpenCV.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Use pretrained YOLO models (YOLOv5/YOLOv7/YOLOv8 or similar) to detect objects in images and videos.	4	LTC	--- NOT APPLICABLE ---
30	Perform inference using pretrained weights and interpret YOLO outputs such as bounding boxes, class labels, and confidence scores.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Analyze detection performance and refine the model by adjusting parameters, augmentations, and training strategies.	4	LTC	--- NOT APPLICABLE ---
30	Train an object detection model on custom data using frameworks like YOLO, TensorFlow, or PyTorch	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	One minute paper

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain key components such as the encoder–decoder structure, skip connections, and feature maps.	3	LTC	--- NOT APPLICABLE ---
30	Implement and train a U-Net model using frameworks like TensorFlow/Keras or PyTorch.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain key components such as the Region Proposal Network (RPN), ROI Align, bounding box regression, and mask prediction branch.	3	LTC	--- NOT APPLICABLE ---
30	Implement and run inference using pretrained Mask R-CNN models to generate bounding boxes, class labels, and segmentation masks.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Build and train deep learning or classical ML models for classifying scenes into predefined categories	4	LTC	--- NOT APPLICABLE ---
30	Evaluate model performance using accuracy, confusion matrix, and other relevant metrics	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Explain key components of a scene graph—objects, attributes, and relationships	3	LTC	--- NOT APPLICABLE ---
30	Apply deep learning–based pipelines for detecting objects and predicting pairwise relationships.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Implement depth estimation using classical algorithms (Block Matching, Semi-Global Matching) or deep learning–based methods.	4	LTC	--- NOT APPLICABLE ---
30	Generate and visualize disparity/depth maps from stereo image pairs	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain key concepts such as camera projection models, epipolar geometry, structure-from-motion (SfM), and multi-view geometry.	4	LTC	--- NOT APPLICABLE ---
30	Reconstruct basic 3D structures using classical pipelines or modern deep learning–based methods.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Apply visual SLAM techniques (e.g., ORB-SLAM, RTAB-Map) for real-time robot localization and environment mapping.	3	LTC	--- NOT APPLICABLE ---
30	Integrate vision-based perception with robotic manipulation for tasks such as object pick-and-place	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Implement real-time inference pipelines using hardware accelerators (GPU, TPU, Jetson devices).	3	LTC	--- NOT APPLICABLE ---
30	Deploy, test, and evaluate an integrated object detection system for stability, speed, and accuracy	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Use annotation tools to label scenes, objects, and attributes for dataset preparation.	3	LTC	--- NOT APPLICABLE ---
30	Prepare complete annotated datasets suitable for training deep learning models for scene recognition.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain how hybrid approaches leverage U-Net's pixel-level accuracy with Mask R-CNN's object-level instance separation.	4	LTC	--- NOT APPLICABLE ---
30	Implement a combined pipeline where U-Net refines masks generated by Mask R-CNN or vice versa.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain how depth maps and segmentation masks can be fused for improved scene interpretation.	4	LTC	--- NOT APPLICABLE ---
30	Implement depth–segmentation fusion pipelines using stereo depth, monocular depth, or LiDAR data.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Apply classical and deep learning–based techniques for lane detection (e.g., Canny, Hough Transform, semantic segmentation).	4	LTC	--- NOT APPLICABLE ---
30	Implement obstacle detection using object detection models (YOLO, SSD, Faster R-CNN) or depth-based methods.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 25

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain key evaluation metrics for classification, detection, and segmentation tasks (Accuracy, Precision, Recall, F1, IoU, mAP, Dice Score, etc.).	4	LTC	--- NOT APPLICABLE ---
30	Analyze model performance using confusion matrices, ROC curves, PR curves, and error analysis.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 26

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Integrate exported models into applications running on edge devices, mobile platforms, or cloud servers.	4	LTC	--- NOT APPLICABLE ---
30	Test and validate the performance, accuracy, and real-time behavior of deployed vision models.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 27

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Extract individual frames from video streams for preprocessing, annotation, or model training.	4	LTC	--- NOT APPLICABLE ---
30	Handle real-time frame capture from webcams or external video sources.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 28

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Apply uniform, random, and interval-based frame sampling techniques to reduce video redundancy.	3	LTC	--- NOT APPLICABLE ---
30	Implement frame selection strategies for action recognition, tracking, and video summarization tasks.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 29

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Understand the process of exporting computer vision models for deployment across different platforms.	3	LTC	--- NOT APPLICABLE ---
30	Test and validate deployed models to ensure performance, accuracy, and real-time capability.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 30

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain how histogram-based techniques can be used to detect significant scene changes.	3	LTC	--- NOT APPLICABLE ---
30	Compute color or intensity histograms for consecutive video frames.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 31

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Build simple and interactive web frontends using Flask for image/video upload and processing.	3	LTC	--- NOT APPLICABLE ---
30	Implement backend logic to handle requests, run inference, and return results (images, bounding boxes, masks, labels).	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 32

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Build and train multi-label classification models using PyTorch with architectures like ResNet, EfficientNet, or custom CNNs.	3	LTC	--- NOT APPLICABLE ---
30	se appropriate loss functions (Binary Cross-Entropy, BCEWithLogitsLoss) and activation functions (Sigmoid) for multi-label outputs.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 33

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Understand the concept of transfer learning and why pretrained CNNs are effective for image classification.	4	LTC	--- NOT APPLICABLE ---
30	Train, validate, and optimize transfer learning models using frameworks like PyTorch or TensorFlow.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 34

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain the assumptions and mathematical foundation of the Lucas–Kanade optical flow method.	3	LTC	--- NOT APPLICABLE ---
30	Implement sparse or dense optical flow estimation using Lucas–Kanade with OpenCV or PyTorch.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 35

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Explain how RL can be applied to computer vision tasks such as object tracking, navigation, and active perception.	3	LTC	--- NOT APPLICABLE ---
30	Implement basic RL algorithms (Q-Learning, Deep Q-Networks) using Python and PyTorch.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 36

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Apply quantization techniques such as dynamic, static, and QAT (Quantization-Aware Training). Optimize and accelerate inference using NVIDIA TensorRT by building optimized engines from ONNX models.	4	LTC	--- NOT APPLICABLE ---
30	Evaluate performance improvements (latency, throughput, resource usage) after quantization and TensorRT optimization.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 37

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Load and fine-tune pretrained Vision Transformer models (e.g., ViT, DeiT, Swin Transformer) using PyTorch.	4	LTC	--- NOT APPLICABLE ---
30	Train, evaluate, and compare Vision Transformer models with CNN baselines for image classification tasks.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 38

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Implement popular SSL frameworks (SimCLR, BYOL, MoCo, MAE) using PyTorch.	4	LTC	--- NOT APPLICABLE ---
30	Generate pseudo-labels through pretext tasks like rotation prediction, patch ordering, or image masking.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 39

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Use MONAI and PyTorch frameworks to build, train, or fine-tune diffusion models for medical image generation (MRI, CT, X-ray, Ultrasound).	3	LTC	--- NOT APPLICABLE ---
30	Generate realistic synthetic medical images for tasks such as augmentation, anomaly simulation, and privacy-preserving dataset expansion.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

Skilling Course DELIVERY Plan:

Skilling session no	Topics/Experiments	CO-Mapping
1	Image I/O- read/write images-. Color Space (RGB ↔ HSV ↔ LAB ↔ Grayscale)	CO4
2	Image Enhancement Techniques (Gaussian, Median, Bilateral)	CO4
3	Image Sharpening & Edge Enhancement	CO4
4	Histogram Skills (Plotting, Equalization, CLAHE)	CO4
5	Thresholding & Basic Segmentation Techniques	CO4
6	Morphological Operations (Erode, Dilate, Open, Close)	CO4
7	Edge Detection Techniques (Sobel, Canny, Prewitt)	CO4
8	Feature Detection Skills (Harris, FAST)	CO4
9	Feature Extraction Skills (SIFT, SURF, ORB)	CO4
10	HOG Feature Extraction & Visualization	CO4
11	Classical ML Pipeline (SVM/KNN + HOG)	CO4
12	Transfer Learning (ResNet/MobileNet)	CO4
13	Data Augmentation Techniques	CO4
14	3D Object Reconstruction using multiple 2D images	CO4
15	PROJECT :Real-Time Vehicle Type Classification using shape and size	CO4
16	Identification and classification of waste and non-waste material in daily life use	CO4
17	Dog Detection in campus area	CO4
18	PROJECT :Underwater image enhancement	CO4
19	PROJECT :Calorie Estimator and Food Image Classification	CO4
20	PROJECT :Detection of helmet on bike riders entering campus	CO4
21	PROJECT :Genre classification of videos based on deep learning approach	CO4
22	PROJECT :Object detection in night vision	CO4
23	PROJECT :Custom Plant disease detection and classification	CO4

Skilling session no	Topics/Experiments	CO-Mapping
24	PROJECT :Detection of optimal empty space for a certain type of vehicle in parking spaces	CO4
25	PROJECT :Explainable Cancer classification from Histopathology	CO4
26	PROJECT :Real-time Video Sequencing & Semantic Summarization.	CO4

Skilling Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain the purpose and characteristics of different color spaces such as RGB, HSV, LAB, and Grayscale.	3	LTC	--- NOT APPLICABLE ---
30	Analyze how color space transformations affect image representation and their applications in tasks like segmentation, detection, and enhancement.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Apply Gaussian filtering for smoothing and reducing Gaussian noise.	3	LTC	--- NOT APPLICABLE ---
30	Apply Bilateral filtering to achieve edge-preserving smoothing for high-quality enhancement.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explain edge enhancement techniques using operators such as Laplacian, Sobel, and Prewitt filters.	3	LTC	--- NOT APPLICABLE ---
30	Apply sharpening methods such as Unsharp Masking and High-Boost filtering.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Apply histogram equalization to enhance global image contrast.	4	LTC	--- NOT APPLICABLE ---
30	Use CLAHE (Contrast Limited Adaptive Histogram Equalization) to perform localized contrast enhancement while avoiding over-amplification.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Understand the fundamentals of image thresholding for separating foreground and background regions.	3	LTC	--- NOT APPLICABLE ---
30	Apply basic thresholding methods such as global thresholding, adaptive thresholding, and Otsu’s method.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Apply compound morphological operations like opening and closing for noise removal and structure refinement.	3	LTC	--- NOT APPLICABLE ---
30	Use structuring elements (kernels) of different shapes and sizes to perform desired morphological effects.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Apply the Canny edge detector, including steps like noise reduction, gradient calculation, non-maximum suppression, and hysteresis thresholding.	4	LTC	--- NOT APPLICABLE ---
30	Implement Canny edge detector, including steps like noise reduction, gradient calculation, non-maximum suppression, and hysteresis thresholding.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Explain the working principles of the Harris Corner Detector and its use in detecting corner-like structures.	2	LTC	--- NOT APPLICABLE ---
30	Apply the FAST (Features from Accelerated Segment Test) algorithm for high-speed keypoint detection.	3	Talk	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Extract robust local descriptors using SIFT and SURF for scale- and rotation-invariant matching.	3	LTC	--- NOT APPLICABLE ---
30	Apply ORB for efficient feature extraction suitable for real-time applications.	3	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	Extract HOG features from images using libraries such as OpenCV or scikit-image.	3	LTC	--- NOT APPLICABLE ---
30	Visualize HOG descriptors to interpret gradient orientations and feature patterns.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
40	rain classical ML classifiers such as Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) using extracted HOG features.	3	LTC	--- NOT APPLICABLE ---
30	Extract discriminative features using Histogram of Oriented Gradients (HOG).	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Load and fine-tune pretrained ResNet and MobileNet models for custom image classification tasks.	3	LTC	--- NOT APPLICABLE ---
30	Modify classifier layers to adapt pretrained networks to new datasets and problem domains.	4	LTC	--- NOT APPLICABLE ---
20	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
40	Explain various augmentation techniques such as flipping, rotation, scaling, cropping, brightness/contrast adjustment, and noise addition.	3	LTC	--- NOT APPLICABLE ---
30	Apply advanced augmentation methods such as Cutout, Mixup, CutMix, and Random Erasing.	3	Talk	--- NOT APPLICABLE ---
20	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
60	Project review	2	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
60	Project review	1	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
60	Project review	2	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
60	Project review	3	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
60	Project review	2	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
				---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
60	Project review	2	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
60	Project review	3	LTC	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
60	Project review	2	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE ---
60	Project review	1	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
60	Project review	2	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
60	Project review	2	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 25

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
				---
60	Project review	2	Chalk	--- NOT APPLICABLE ---
30	Evaluation & Viva	3	LTC	--- NOT APPLICABLE ---

SESSION NUMBER : 26

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Recap & Attendance	2	Talk	--- NOT APPLICABLE ---
60	Project review	2	Talk	--- NOT APPLICABLE ---
30	Evaluation & Viva	2	Talk	--- NOT APPLICABLE ---

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDEDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	--	--	---	---	---	---	---	---	---
	Tutorial	--	--	---	---	---	---	---	---	---
	Lab	H-S1	H-S1	---	---	---	---	---	---	---
	Skilling	--	--	---	---	---	---	---	---	---
Tue	Theory	---	---	---	--	--	---	---	---	---
	Tutorial	---	---	---	--	--	---	---	---	---
	Lab	---	---	---	H-S1	H-S1	---	---	---	---
	Skilling	---	---	---	--	--	---	---	---	---
Wed	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Thu	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--

	Hour	1	2	3	4	5	6	7	8	9
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Fri	Theory	--	--	---	---	---	---	---	---	---
	Tutorial	--	--	---	---	---	---	---	---	---
	Lab	--	--	---	---	---	---	---	---	---
	Skilling	H-S1	H-S1	---	---	---	---	---	---	---
Sat	Theory	---	---	---	---	---	---	---	---	---
	Tutorial	---	---	---	---	---	---	---	---	---
	Lab	---	---	---	---	---	---	---	---	---
	Skilling	---	---	---	---	---	---	---	---	---
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCS

DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCS

EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4
End Semester Summative Evaluation Total= 40 %	**Lab End Semester Exam**	Weightage	15		100	5	5	5	
		Max Marks	75			25	25	25	
	SEM End Project	Weightage	25		100				25
		Max Marks	50						50
In Semester Formative Evaluation Total= 30 %	**Global Challenges**	Weightage	6		110				6
		Max Marks	50						50
	Skilling Continuous Evaluation	Weightage	8		100				8
		Max Marks	50						50
	MOOCs Review	Weightage	8		100				8

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4
		Max Marks	50						50
	Continuous Evaluation - Lab Exercise	Weightage	8		110	2.5	2.5	3	
		Max Marks	75			25	25	25	
In Semester Summative Evaluation Total= 30 %	**Skill In-Sem Exam**	Weightage	15		100				15
		Max Marks	50						50
	Lab In Semester Exam	Weightage	15		100	5	5	5	
		Max Marks	75			25	25	25	

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course
In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
SUKLA SATAPATHY	P	1-A	-	-	-	-
SUKLA SATAPATHY	S	1-A	-	-	-	-
Harsha P	P	1-B	-	-	-	-
Harsha P	S	1-B	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(Sasidhar Kothuru)(7246)(CSE-1)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of AI&DS

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS
(Sign with Office Seal)
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